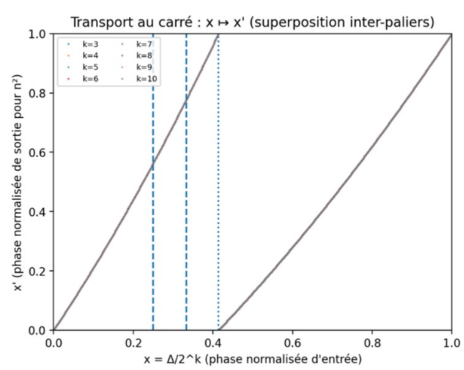
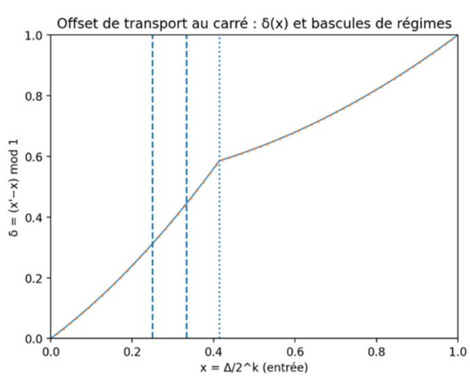
Towards a Geometric Elucidation of the Impossibility of Squaring the Circle — and What It Says

For more than two millennia, squaring the circle has embodied the archetype of an insoluble problem. Inherited from Greek geometry and reformulated over the centuries, it reached its formal closure in 1882 with Lindemann's proof of the transcendence of π: it is impossible, using only straightedge and compass, to construct a square with the same area as a given circle, since $\sqrt{\pi}$ is not a constructible number. Yet behind this formal impossibility lies a deeper question, one that the classical metric approach has obscured: why do the circle and the square seem incapable of truly "speaking" to one another? Why does every attempt to make their measures coincide fail, not because of insufficient precision, but because of a more fundamental structural incompatibility?

What we see from last study between product on dyacic's level 6 and 12 become enough clear.



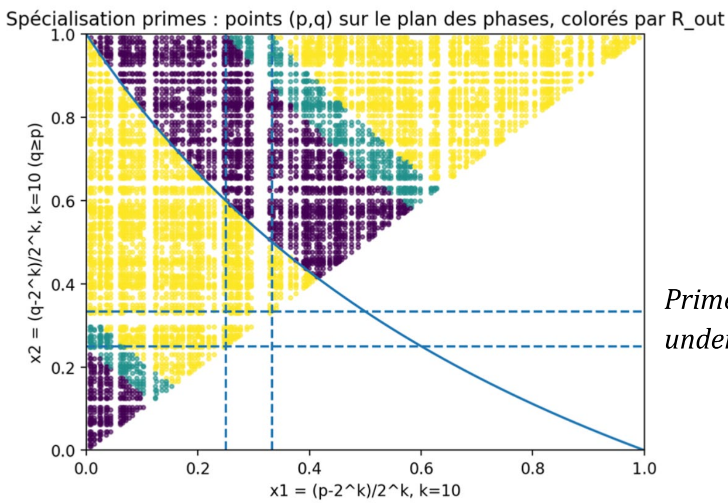
Phase transport induced by squaring on dyadic power circles, showing scale-independent regime changes.



Transport offset as the primary observable, highlighting regime transitions.

This work does not claim to circumvent Lindemann's impossibility, nor to propose a forbidden geometric construction. It proposes instead a radical reformulation of the problem: squaring the circle is not merely a matter of equating areas, but a matter of transporting information between two irreducibly different mathematical languages.

This investigation arises from an in-depth study of prime and semiprime numbers, no longer regarded as simple arithmetic objects, but as angular positions transported on dyadic power circles. By exploring the dynamics of products, squares, and roots across these concentric circles, we reveal a geometric structure that is stable across scales, yet invisible within the classical algebraic framework.



Prime pairs sampling the underlying transport structure.

This framework shows that squaring and extracting a square root do not preserve the same geometric information: the circle carries phase and regime invariants that squaring, as a projective operator, partially erases. This structural dissymmetry sheds new light on the impossibility of squaring the circle—not as a technical limitation tied to π, but as the consequence of an irreducible change of geometric support between the algebraic line and power circles.

Thus, squaring the circle appears less as an isolated problem than as a deep signal about how the Real is represented: certain transformations require richer transport spaces than those allowed by classical constructions. This shift in perspective opens a renewed geometric reading of fundamental operations and invites us to rethink the relationship between number, form, and mathematical reality.

The circle conveys a continuous phase, a smooth orientation, an angular memory without rupture or loss. It is the natural support of continuous, oriented information.

The square—or more generally exponentiation and arithmetic products—acts through folds, renormalizations (binary carries), regime changes, and discrete stratifications. It redistributes information across concentric dyadic crowns, with internal boundaries and dynamical bifurcations.

By representing natural integers not on a line but on concentric dyadic circles (mantissa levels in base 2, with an angular offset θ encoding relative phase), this tension becomes visible. Squaring then becomes a transport operator between circles: a quadratic law on the phase, folded by a carry once a critical threshold (~0.414) is crossed. The product of two numbers imposes compatibility between two input phases and a resulting phase, delimited by hyperbolic carry boundaries that organize the space of admissibility.

What this framework reveals is that the historical failure of squaring the circle stems from seeking a scalar equality where one should recognize an incompatibility of transports. The circle preserves phase; the square folds, renormalizes, and stratifies it. Neither can fully absorb the logic of the other without irreversible information loss.

Mathematical formulation of the transport laws

Let $k \in \mathbb{N}$and let an integer $n \in [2^k, 2^{k+1})$be represented by its normalized dyadic phase

$$x = \frac{n - 2^k}{2^k} \in [0,1).$$

Squaring transport

The squaring operation $n \mapsto n^2$induces a phase transport $x \mapsto x'$between dyadic power circles given by the piecewise law

$$\boxed{x' = \begin{cases} 2x, & \text{if } x < \tau, \\ 2x - 1, & \text{if } x \geq \tau, \end{cases} \tau = \sqrt{2} - 1.}$$

The associated transport offset,

$$\boxed{\delta(x) = x' - x (\mathrm{mod} 1),}$$

is a continuous but non-linear observable exhibiting clear regime transitions at the dyadic carry threshold $x = \tau$.

Multiplicative transport

For a product $n = pq$, with

$$x_1 = \frac{p - 2^k}{2^k}, x_2 = \frac{q - 2^k}{2^k},$$

the output phase satisfies

$$\boxed{x' = x_1 + x_2 + \varepsilon(x_1, x_2)(\mathrm{mod}1),}$$

where the correction term $\varepsilon$is determined by the dyadic carry condition

$$(1 + x_1)(1 + x_2) = 2,$$

which partitions the phase plane into distinct transport regimes.

A symmetric residual offset may be defined as

$$\boxed{\delta_m = x' - \frac{x_1 + x_2}{2}(\mathrm{mod}1),}$$

providing a measurable quantity that captures the geometric deformation induced by multiplicative transport.

Far from being a speculative exercise, this dyadic descent—observable from the smallest levels (k = 3 to 10) and stable across scales—highlights structuring phenomena: transport regimes (R1, R2, R3), transport offsets Δθ as central observables, domains of unique versus ambiguous resolution, and regions of low entropy (readable zones for inversion). These structures, present at small scale, amplify macroscopically and may illuminate behaviors long attributed to randomness or brute complexity (such as the factorization of large semiprimes).

Understood this way, squaring the circle ceases to be a technical prohibition. It becomes the visible symptom of a deeper conceptual misunderstanding: the profound non-commensurability between oriented continuity and stratified discreteness. And it is precisely in the articulation—rather than the mutual reduction—of these two languages that a richer understanding may reside.

This text is therefore not a “solution” in the classical sense; it is an elucidation, and potentially a signal: the dyadic-concentric geometry of powers and products may open a new arithmetico-dynamic grammar, in which the carry plays the role of a singular operator, and phase becomes the observable—the structure itself—that speaks when values alone fall silent.

Author: Daniel CICCIA
Date: 09 February 2026